

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6B 11—COMPLEX ANALYSIS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 11—COMPLEX ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following function $f(z)$, of the complex variable z , is not analytic at all the points of the complex plane ?

(A) $f(z) = z^2$. (B) $f(z) = e^z$.
(C) $f(z) = \sin z$. (D) $f(z) = \log z$.

2. If z is a non-zero complex number, then for $n = 1, 2, 3, \dots$, $z^{1/n}$ is :

(A) $\exp(n \log z)$. (B) $\exp\left(\frac{1}{n} \log z\right)$.
(C) $\exp\left(\frac{1}{n} \log \frac{1}{z}\right)$. (D) $\exp\left(n \log \frac{1}{z}\right)$.

3. Real part of the function $|z|^2$ is :

(A) $x^2 - y^2$. (B) $2xy$.
(C) $x^2 + y^2$. (D) $y^2 - x^2$.

4. If a function f is analytic throughout a simple connected domain D , then $\int_C f(z) dz = \text{_____}$.

(A) 0. (B) $2\pi i$.
(C) $2\pi i f(z)$. (D) 1.

5. The integral of $\int_C \frac{dz}{z-i}$ where C is the circle $|z| = 2$ is :

(A) $2\pi i$. (B) $-\pi i$.
(C) πi . (D) $-2\pi i$.

6. The integral of the function $\int_C \frac{e^z}{z} dz$, where C is the unit circle is :

(A) $2\pi i$. (B) $-\pi i$.
(C) πi . (D) $-2\pi i$.

7. The integral of the function $\int_C \frac{z+2}{z-2} dz$ where C is the unit circle is :

(A) 0. (B) 1.
(C) -1. (D) π .

8. If f is continuous in a domain D and if $\int_C f(z) dz = 0$ for every simple closed positively oriented contour C in D , then :
- (A) f is analytic in D . (B) f is real valued in D .
 (C) f is constant in D . (D) f is imaginary in D .
9. Piecewise smooth curve is also known as :
- (A) Contour. (B) Smooth curve.
 (C) Circle. (D) Regular curve.
10. The radius of convergence of the series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ is :
- (A) 1. (B) 0.
 (C) -1. (D) ∞ .
11. The center of the power series $\sum_{n=0}^{\infty} (z + 4i)^n$ is :
- (A) $4i$. (B) $2i$.
 (C) 4. (D) $-4i$.
12. The zero of the function $\frac{z}{\cos z}$ is :
- (A) 1. (B) 0.
 (C) -1. (D) π .
13. The principal part of $f(z)$ at z_0 consists of infinite number of terms, then z_0 is known as :
- (A) Pole. (B) Essential singular point.
 (C) Removable singular point. (D) Simple pole.
14. The power series $b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots$ converges.
- (A) Inside of some circle $|z| = R$. (B) On the circle $|z| = 1$.
 (C) On some circle $|z| = R$. (D) Outside of some circle $|z| = R$.

15. The residue of the function $\frac{z^3}{(z-1)^4(z-2)(z-3)}$ at $z = 3$ is :

(A) -8 .

(B) $\frac{101}{16}$.

(C) 0 .

(D) $\frac{27}{16}$.

16. Residue of the function $\frac{4}{1-z}$ at the singular points is :

(A) 4 .

(B) -4 .

(C) 2 .

(D) -2 .

17. Given $f(z) = \frac{z^2}{z^2 + a^2}$. Then :

(A) $z = ia$ is a pole and $\frac{ia}{2}$ is a residue at $z = ia$ of $f(z)$.

(B) $z = ia$ is a simple pole and ia is a residue at $z = ia$ of $f(z)$.

(C) $z = ia$ is a simple pole and $-\frac{ia}{2}$ is a residue at $z = ia$ of $f(z)$.

(D) None of the above.

18. The value of $\oint \frac{1}{z^2} dz$, where the contour is the unit circle traversed clockwise is :

(A) $-2\pi i$.

(B) 0 .

(C) $2\pi i$.

(D) $4\pi i$.

19. The integral of $\int_C \frac{1}{z \sin z} dz$ where C is the unit circle oriented in the positive direction is :

(A) πi .

(B) $2\pi i$.

(C) $-\pi i$.

(D) 0 .

20. The zero of order is known as :

(A) Complex zero.

(B) Simple zero.

(C) Singularity.

(D) None of these.

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Define holomorphic function in a domain D . And give an example for an entire function.
2. Prove or disprove : if f is differentiable at a point z_0 , then f is continuous at that point.
3. Define harmonic function with example.
4. Prove that $\sin^2 z + \cos^2 z = 1$.
5. State ML inequality.
6. Define the path independence for a contour integral.
7. State maximum modulus theorem.
8. Prove that $\int_a^b f(t) dt = - \int_b^a f(t) dt$.
9. Prove or disprove if $\lim_{n \rightarrow \infty} z_n = 0$, then $\sum_{k=1}^{\infty} z_k$ converges.
10. Find the radius of convergence of $\sum_{k=1}^{\infty} \frac{z^k}{k}$.
11. Define pole of order n . Give an example of a function with simple pole at $z = 1$.
12. Find the principal part in the Laurent series expansion about the origin of the function $f(z) = \frac{\sin z}{z^4}$.

Turn over

13. State Rouché's theorem.
14. Find the residue of $\frac{\sin z}{z}$ at $z = 0$.
15. How many zeroes are in the disc $|z| = 1$ for the function $f(z) = z^9 - 8z^2 + 5$.

(10 × 3 = 30 marks)

Section B

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Check whether the function U is harmonic or not if so find its harmonic conjugate $U(x, y) = x^3 - 3xy^2 - 5y$.
17. Find all the solutions of the equation $\sin z = 5$.
18. State and prove Fundamental theorem of algebra.
19. State and prove Morera's theorem.
20. Find the Taylor's series expansion with centre $z_0 = 2i$ of $f(z) = \frac{1}{1-z}$.
21. Identify the singular points and classify them $f(z) = \frac{\sin ze^{\left(\frac{1}{z-1}\right)}}{z(1+z)}$.
22. Find residue of $e^{\left(\frac{1}{z}\right)}$ at $z = 0$.
23. Find $\int_{-\infty}^{\infty} \frac{1}{x^4 + 1} dx$.

(5 × 6 = 30 marks)

Section C (Essay Questions)

Answer any two questions.

Each question carries 10 marks.

24. State and prove Cauchy Riemann Equation. Also state the sufficient condition for differentiability.
25. State and prove Cauchy's integral formula for derivatives.
26. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z-2| < 2$.
27. State and prove Cauchy's residue theorem.

(2 × 10 = 20 marks)